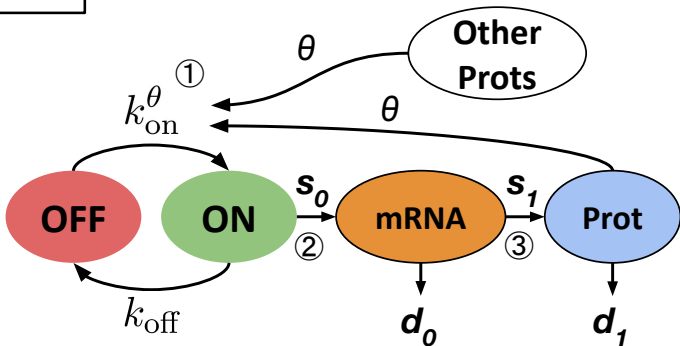


A Model and hypotheses

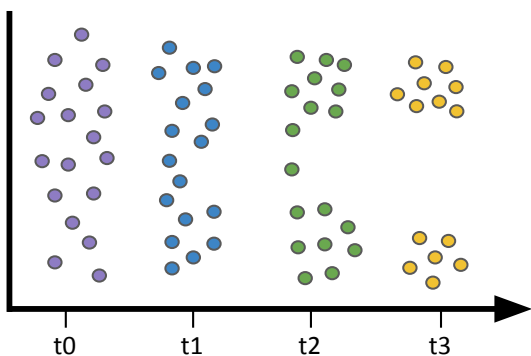


Two-state gene expression model with :

- 1) $k_{\text{on}}^{\theta}(P_{\text{global}}) = k_0 + (k_1 - k_0)f(P_{\text{global}}, \theta)$
- 2) $M(t) \xrightarrow{k_{\text{on}}^{\theta}(P_{\text{global}}(t))} M(t) + \mathcal{E}\left(\frac{k_{\text{off}}}{s_0}\right), \quad \frac{dM}{dt} = -d_0 M$
- 3) $\frac{dP}{dt} = s_1 M - d_1 P$

B CardamomOT input data

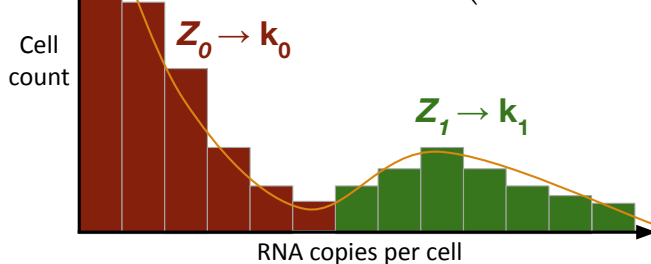
Sequence of scRNA-seq snapshots



C Preprocessing

Calibration of a Negative-Binomial mixture model for each gene and timepoint with :

$$M(t) | P_{\text{global}}(t) \sim \text{NB}\left(\frac{k_{\text{on}}^{\theta}(P_{\text{global}}(t))}{d_0}, \frac{k_{\text{off}}}{s_0}\right)$$



D Inference refining cycle

Step 1: Update protein trajectories using Optimal Transport with GRN-based cost :

1.1) For every cell-cell coupling, we find the potential $P_{c_{t_j}, c'}$:

$M_{c_{t_j}, c'}$ Implies $P_{c_{t_j}, c'}$

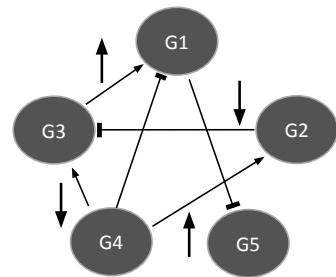
1.2) We update the GRN-based cost, and using $P_{c_{t_j}}$ and $P_{c_{t_j}, c'}$, $P_{c'}$ is sampled following $OT_C^j(\hat{P}_{t_j}, \hat{P}_{t_{j+1}})$:

$$C_{c_{t_j}, c'}^j = \text{loss}\left(k_{\text{on}}^{\theta}\left(P_{c_{t_j}, c'}(t_{j+1})\right), k_{z_{c'}}\right)$$

$$OT_C^j(\hat{P}_{t_j}, \hat{P}_{t_{j+1}}) = \arg \min_{\pi} \left\{ \sum_{c \in \hat{P}_{t_j}} \sum_{c' \in \hat{P}_{t_{j+1}}} C_{c, c'}^j \pi(c, c') + \text{Reg} \right\}$$

Step 2: Update the Gene Regulatory Network following :

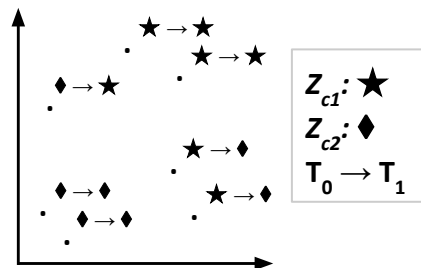
$$\theta^* = \arg \min_{\theta} \sum_{c \in \hat{P}} \text{loss}(k_{\text{on}}^{\theta}(P_c), k_{z_c}) + \text{Reg}$$



Step 3: Update cell basin labels

$$\gamma^* = \arg \min_{\gamma} \left\{ \sum_{z \in Z} \sum_{c \in \hat{P}} - \left(\ln(\text{Pr}_c(z)) + \varepsilon \ln \left(1 - \text{loss}(k_{\text{on}}^{\theta}(P_c), k_{z_c}) \right) \right) \gamma(c, z) \right\}$$

such that : $\sum_{c \in \hat{P}} \gamma(c, z) = \sum_{c \in \hat{P}} \text{Pr}_c(z)$, and $\sum_{z \in Z} \gamma(c, z) = 1$. Cells are assigned $z_c = \arg \max_z \gamma^*(c, z)$



E Simulation of new trajectories

Parameters from trained model :

$$\theta, d_0, d_1, \frac{k_{\text{off}}}{s_0}, M(t_0) \dots$$

Parameter perturbation (e.g. In-silico K.O.)

Harissa
based simulation module

Simulation of alternative trajectories

CardamomOT output

